

Md Amit Hasan Arovi

Assistant Teaching Professor, Computer Science and Engineering
The New Mexico Institute of Mining and Technology (New Mexico Tech), Socorro, NM
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Research Interests

Computer Systems; Concurrent and Parallel Computing; Non-Blocking Algorithms and Data Structures; Safe Memory Reclamation (SMR); Recyclable Data Structures; Optimistic and Wait-Free Traversals; Correctness and Progress Guarantees; High-Performance Multicore Systems.

Education

Ph.D., Computer Science and Engineering Summer 2026
The Pennsylvania State University, University Park, PA
Advisor: Dr. Ruslan Nikolaev
Dissertation: *Bridging the Gap Between Safe Memory Management and Non-Blocking Data Structures*

M.S., Computer Science and Engineering Spring 2025
The Pennsylvania State University, University Park, PA
Advisor: Dr. Ruslan Nikolaev

B.S., Computer Science and Engineering Dec 2011 – Nov 2015
Islamic University of Technology, Bangladesh
Thesis: *Speech-Based Interaction for the Visually Impaired*
Advisor: Dr. Md Kamrul Hasan

Academic Appointment

Assistant Teaching Professor Aug 2026 – Present
Department of Computer Science and Engineering
The New Mexico Institute of Mining and Technology (New Mexico Tech), Socorro, NM

Peer-Reviewed Publications

PLDI 2025:

RRR-SMR: Reduce, Reuse, Recycle: Better Methods for Practical Lock-Free Data Structures

Md Amit Hasan Arovi, Ruslan Nikolaev.

Proceedings of the 46th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI).

DOI: [10.1145/3729337](https://doi.org/10.1145/3729337)

PPoPP 2026:

Fixing Non-Blocking Data Structures for Better Compatibility with Memory Reclamation Schemes

Md Amit Hasan Arovi, Ruslan Nikolaev.

Proceedings of the 31st ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (PPoPP).

DOI: [10.1145/3774934.3786455](https://doi.org/10.1145/3774934.3786455)

SPAA 2025:

Brief Announcement: SCOT: Fix Non-Blocking Data Structures, Not Memory Reclamation

Md Amit Hasan Arovi, Ruslan Nikolaev.

Proceedings of the 37th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA).

DOI: [10.1145/3694906.3743348](https://doi.org/10.1145/3694906.3743348)

SPAA 2026:

Brief Announcement: Recyclable Optimistic-Traversal Data Structures

Md Amit Hasan Arovi, Ruslan Nikolaev.

Proceedings of the 38th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA).

DOI: [10.1145/3816782.3819211](https://doi.org/10.1145/3816782.3819211)

ICCIT 2015:

Speech Based Text Correction Tool for the Visually Impaired

Md. Nafiz Hasan Khan; **Md Amit Hasan Arovi**; Hasan Mahmud; Md. Kamrul Hasan; Husne Ara Rubaiyeat.

Proceedings of the 18th International Conference on Computer and Information Technology (ICCIT), IEEE.

pp. 150–155. Published: December 2015.

DOI: [10.1109/ICCITech.2015.7488059](https://doi.org/10.1109/ICCITech.2015.7488059)

Research Experience

Graduate Research Assistant, The Pennsylvania State University

Research appointments: Fall 2022, Summer 2023, Spring 2024, Summer 2024, Fall 2024, Summer 2025, Fall 2025, Spring 2026, Summer 2026

- Developed RRR-SMR, a methodology for copy-free node reuse and transfer across recyclable lock-free queues, linked lists, and Natarajan–Mittal trees while addressing ABA safety, two-phase deletion, and explicit node ownership.
- Developed SCOT, a data-structure adaptation technique that enables safe optimistic traversals under robust SMR schemes, including hazard pointers, hazard eras, interval-based reclamation, and Hyaline, without changing the SMR API.
- Established safety, bounded-memory, and progress arguments for the proposed algorithms, including lock-free updates and a helping-based wait-free search extension for linked lists.
- Integrated RRR-SMR and SCOT into R-SCOT, using differentiated tag increments to preserve local recovery and efficient optimistic traversal in recyclable data structures.
- Built and maintained reproducible C and C++ benchmarks and research artifacts; evaluated throughput and memory efficiency on a 128-core AMD EPYC system across multiple SMR schemes, workloads, data structures, and payload sizes.

Teaching Experience

Instructor of Record, The New Mexico Institute of Mining and Technology

- **CSCI 2230: Assembly Language and Machine Organization** – Fall 2026
- **CSCI 1210/1210L: Programming Fundamentals in C and Lab** – Fall 2026
- **CSCI 1001/1001L: Introduction to Computer Science and Information Technology** – Fall 2026

Responsibilities include designing and delivering lectures and lab activities, preparing programming assignments and assessments, holding office hours, coordinating with assigned teaching assistants, and supporting students with varied levels of prior programming and computing experience.

Teaching Assistant, The Pennsylvania State University

- **CMPSC 311: Introduction to Systems Programming** – Spring 2023, Spring 2025
- **CMPSC 473: Operating Systems Design and Construction** – Fall 2023

Responsibilities included leading recitations, holding office hours, grading exams and assignments, and mentoring students through challenging programming projects in C and C++, Unix systems programming, memory management, concurrency, and operating systems.

Conference Presentations

RRR-SMR: Reduce, Reuse, Recycle: Better Methods for Practical Lock-Free Data Structures

Conference presentation, PLDI 2025, Seoul, South Korea.

Slides: [PLDI 2025 presentation on RRR-SMR](#)

Fixing Non-Blocking Data Structures for Better Compatibility with Memory Reclamation Schemes

Conference presentation, PPOPP 2026, Sydney, NSW, Australia.

Slides: [PPOPP 2026 presentation on SCOT](#)

Software and Research Artifacts

RRR-SMR Artifact (PLDI 2025)

Zenodo DOI: [10.5281/zenodo.15258497](https://doi.org/10.5281/zenodo.15258497)

Code: [GitHub](#)

SCOT Artifact (PPOPP 2026)

Zenodo DOI: [10.5281/zenodo.17707898](https://doi.org/10.5281/zenodo.17707898)

Code: [GitHub](#)

R-SCOT Code (SPAA 2026)

Code: [GitHub](#)

Fellowships and Competitive Travel Awards

- **Harry G. Miller Fellowship in Engineering (\$1,000).**
Penn State College of Engineering merit scholarship to attend and present at PPOPP 2026.
- **ACM SIGHPC Student Travel Grant for PPOPP 2026 (\$1,600).**
Travel support to attend and present at PPOPP 2026.
- **PLMW @ PLDI 2025 Travel Scholarship, ACM SIGPLAN (\$2,700).**
Travel support to attend PLMW @ PLDI 2025.
- **ACM SIGPLAN PAC Travel Support for PLDI 2025 (\$1,500).**
Travel support to attend and present at PLDI 2025.
- **ACM SIGACT/SIGARCH Student Travel Grant for SPAA 2025 (\$750).**
Travel support to attend and present at SPAA 2025.

Honors and Awards

Microsoft Imagine Cup (Bangladesh), 2015 – Winner;
World Semifinalist (World Citizenship Track).
Project: **Eye Pointer** – [Demo video \(YouTube\)](#).

Industry Experience

Nilavo Technologies Ltd., Bangladesh

Senior Software Engineer	Jan 2022 – Jul 2022
Software Engineer	Jan 2019 – Dec 2021
Junior Software Engineer	Dec 2015 – Dec 2018

- Built full-stack ASP.NET applications using C#, Microsoft SQL Server, and JavaScript, from specification through deployment.
- Performed debugging, performance optimization, maintenance, and secure-code audits for production systems.